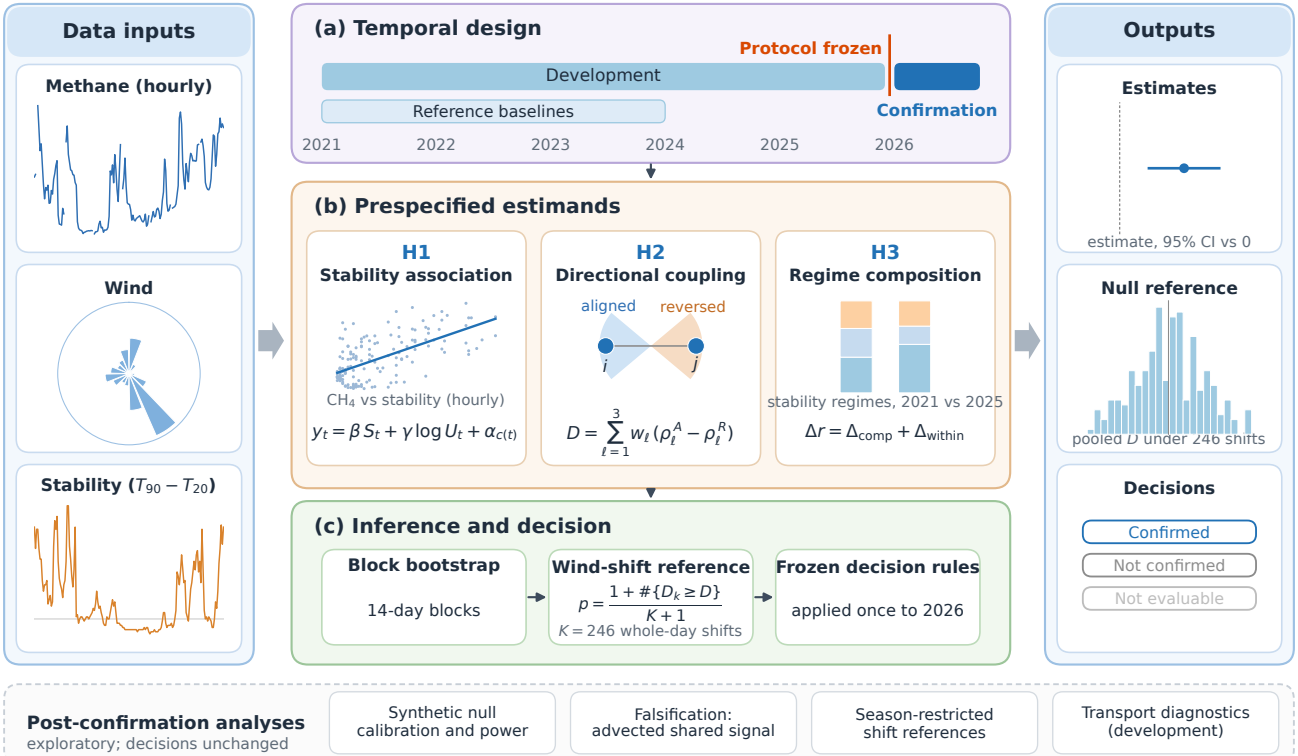

Data inputs
Methane (hourly)
Wind
Stability ($T_{90} - T_{20}$)
(a) Temporal design
Protocol frozen
Development
Reference baselines
Confirmation
2021
2022
2023
2024
2025
2026
(b) Prespecified estimands
H1
Stability association
CH_4 vs stability (hourly)
$y_t = \beta S_t + \gamma \log U_t + \alpha_{c(t)}$
H2
Directional coupling
aligned
reversed
i
j
$D = \sum_{\ell=1}^{3} w_\ell (\rho_\ell^A - \rho_\ell^R)$
H3
Regime composition
stability regimes, 2021 vs 2025
$\Delta r = \Delta_{\text{comp}} + \Delta_{\text{within}}$
(c) Inference and decision
Block bootstrap
14-day blocks
Wind-shift reference
$p = \frac{1 + \#\{D_k \geq D\}}{K + 1}$
$K = 246$ whole-day shifts
Frozen decision rules
applied once to 2026
Outputs
Estimates
estimate, 95% CI vs 0
Null reference
pooled D under 246 shifts
Decisions
Confirmed
Not confirmed
Not evaluable
Post-confirmation analyses
exploratory; decisions unchanged
Synthetic null calibration and power
Falsification: advected shared signal
Season-restricted shift references
Transport diagnostics (development)